LESSON NOTE

ON

(EEE 437)

ELECTRONIC DRAFTING (DESIGN)

GOAL:

The Course is designed to provide the student with knowledge and Skill in design & drafting of electronic Circuits.

OBJECTIVES:

1- Know / Identify the symbols of Components used in electronics.
2- Know how to draw Schematic diagrams
3- Know how to draw block & logic diagrams
4- Know Construction of a dependable & easy-to trouble-shoot prototype or modification ect.
5- Know the production of Prints Circuit boards
6- Know how to draw Wiring Assembly Diagrams
7- Understand the Various types of design & layout of Communication Systems.

MEANING OF BOTH DESIGN & DRAFTING

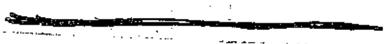
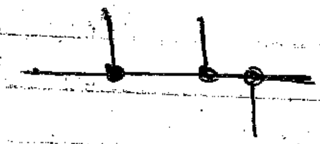
- DESIGN: This is simply the Underlying planning purpose that lays the basis for the making of every device or a System. The person designing is called a DESIGNER.

DESIGNING: Often requires a designer to Consider the aesthetic, functional and Other aspects of an Object or a process, which usually requires Considerable research, thought, modelling, Interactive adjustment and re-design.

DRAFTING: Is the Process of Producing engineering drawing and the skill of producing them is often referred to as technical drawing or drafting. NB: *Design & Drafting not be done by anyone not familiar with all the Electronics Code & standards needed for the design

1.0 → COMMONLY USED ELECTRONIC SYMBOLS

1.1 WIRES AND CONNECTIONS:

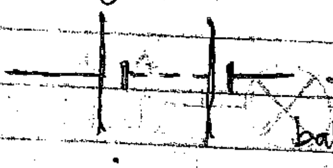
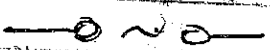
COMPONENT	CIRCUIT SYMBOL	FUNCTIONS
- WIRE		- To pass Current very easily from one part of a cct to another.
- WIRES JOINED		- A "blob" should be drawn where wires are Connected (Joined), but it is Sometimes Omitted. wired Connected at Crossroads should be staggered slightly to form 2 T-Junctions, as shown on the right.
- WIRES NOT JOINED		In Complex diagrams It is often necessary to

draw wires Crossing even though they are not Connected. It is prefer the "bridge" symbol shown on the right because the simple Crossing on the left may be misread as a Join where you have forgotten to add a blob 1.

POWER SUPPLIES:

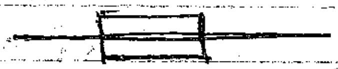
COMPONENT	CIRCUIT SYMBOL	FUNCTIONS
- CELL		- It supplies electrical energy The larger terminal (on the left) is

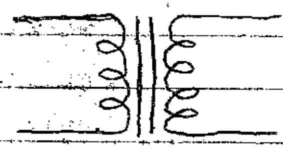
positive (+). A single Cell is often called a battery but strictly a battery is two or more cells joined together.

- BATTERY		- Supplies electrical energy. A battery is more than one Cell. - The larger terminal (on the left) act as +ve
- DC SUPPLY		Supplies electrical energy DC = Direct Current, always flowing in one direction.

COMPONENT — CIRCUIT SYMBOL — FUNCTIONS

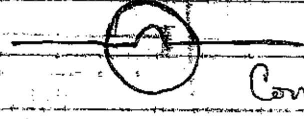
- AC SUPPLY  Supplies electrical energy.
  AC = Alternating Current, continually changing direction.

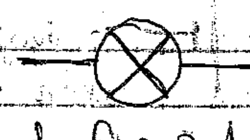
- FUSE  This is a safety device which will 'blow' (melt) if the current flowing via it exceeds a specified value.

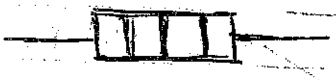
- TRANSFORMER  Two coils of wire linked by an iron core. Transformers are used to step-up (increase) and step-down (decrease) AC voltages. Energy is transferred between the coils by the magnetic field in the core. There is no electrical connection b/w the coils.

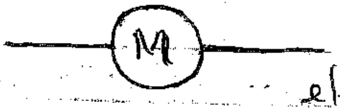
- EARTH :  - This is simply a connection to earth. For many electronic ccts this is the 0V (zero volts) of the power supply, but for mains electricity and some radio circuits it really means the earth. It is also known as "GROUND".

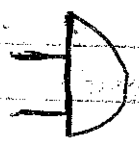
OUTPUT DEVICES : SYMBOLS — FUNCTIONS

- LAMP LIGHTING  This is a transducer which converts electrical energy to light. The symbol is used for a lamp providing illumination, for example a car headlamp or torch bulb.

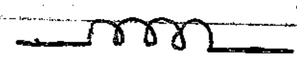
- LAMP (INDICATOR)  - A transducer which converts electrical energy to light. This symbol is used for a lamp which is an indicator, for example a warning light on a car dashboard.

- HEATER  This is a transducer which Converts electrical energy to heat.

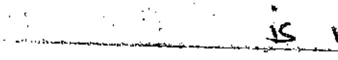
- Motor :  A transducer which Converts electrical energy to Kinetic energy (motion).

- BELL  This is a transducer which Converts electrical energy to Sound.

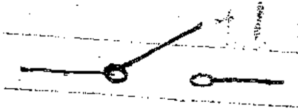
- BUZZER  - A transducer which Converts electrical energy to Sound.

- INDUCTOR (COIL, SOLENOID)  A coil of wire which Creates a Magnetic field when Current passes through it. It may have an iron core inside the coil. It can be used as a transducer converting electrical energy to mechanical energy by pulling on something.

- SWITCHES :

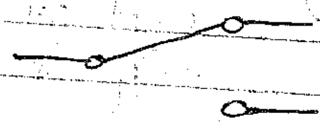
COMPONENT	Cct SYMBOL	FUNCTIONS
- PUSH SWITCH (push-to-make)		A push switch allows current to flow only when the button is pressed. This is the switch used to operate a doorbell.
- PUSH-TO-BREAK SWITCH.		- This type of Push Switch is normally closed (on), it is open (off) only when the button is pressed.

ON-OFF-SWITCH
(SPST)



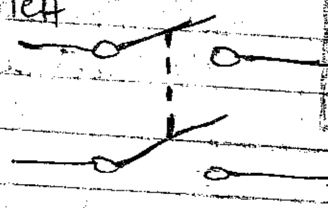
SPST = Single Pole, Single Throw.
An on-off switch allows current to flow only when it is in the close (on) position.

- 2-WAY-SWITCH
(SPDT)



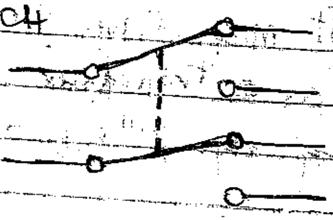
SPDT = Single Pole, Double Throw.
A 2-way changeover switch directs the flow of current to one of two routes according to it's position. Some SPDT switches have a central off position and are described as "on-off-on"

- DUAL ON-OFF SWITCH
(DPST)



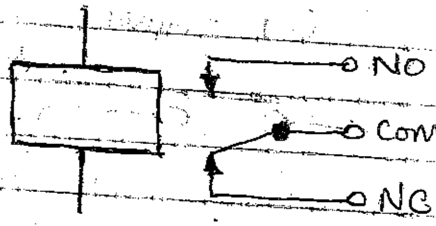
DPST = Double Pole, Single throw
A Dual on-off switch is often used to switch mains electricity because it can isolate both the live and neutral connections.

- REVERSING SWITCH
(DPDT)



DPDT = Double Pole, Double throw.
This switch can be wired up as a reversing switch for a motor. Some DPDT switches have a central off position.

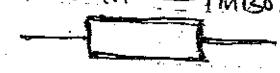
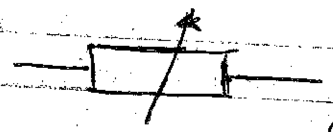
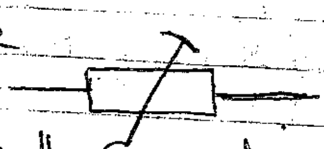
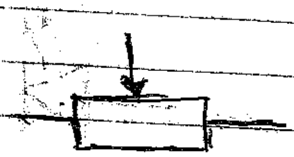
- RELAY



An electrically operated switch, for example a 9v battery cet connected to the coil can switch a 230V AC Mains circuit,

N/B:
NO = Normally Open
COM = Common
NC = Normally closed.

RESISTORS :

COMPONENT	CIRCUIT SYMBOL	FUNCTION
- NORMAL RESISTOR		A resistor restricts the flow of Current, for example to limit the Current passing through an LED. A resistor is used with a Capacitor in a timing Circuit. Some Publications still use the Old resistor symbol : 
- VARIABLE RESISTOR (RHEOSTAT)		This Type of variable resistor with 2 Contacts (a rheostat) is usually used to Control Current. Examples include; adjusting lamp brightness, adjusting motor speed, and adjusting the rate of flow of Charge into a Capacitor in a Timing Circuit.
- VARIABLE RESISTOR (PRESET)		This Type of variable resistor (a preset) is operated with a small Screwdriver or Similar tool. It is designed to be set when the cct is made & then left without further adjustment. Presets are cheaper than normal variable resistors so they are often used in Projects to reduce the Cost.
- VARIABLE RESISTOR (POTENTIOMETER)		This Type of variable resistor with 3 Contacts (a potentiometer) is usually used to Control voltage. It Can be used like this as a transducer Converting position (angle of the Control Spindle) to an electrical Signal.

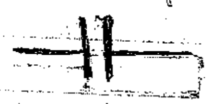
... are electrical devices that stores energy in ... electrical field are ... [illegible]

TYPES - Ceramic
- Electrolytic
- Film
- Silver mica
- Piezoelectric
- Polarised
Non-polarised

- Capacitors are divided into two mechanical groups
- Fixed capacitors with fixed capacitance values & variable capacitors with variable (trimmer) or adjustable (tunable) capacitance values. The most important group is the fixed capacitors. Many got their names from the dielectric

CAPACITOR:

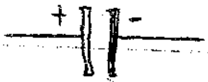
- (NORMAL CAPACITOR)



A Capacitor stores electric charge. A Capacitor is used with a resistor in a timing circuit. It can also be used as a filter, to block DC signals but pass AC signals.

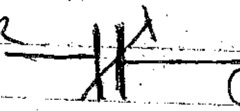
- Capacitors also have functions to keep the voltage at a certain level i.e. keeping the voltage at the same level
- AC flowing but not DC
- Removing noise

- CAPACITOR (POLARISED)



A Capacitor stores electr. charge. This type must be connec the correct way round. A capacitor is used with a resistor in a timing Cct. It can also be used as filt to block DC signals but pass AC signals.

- TRIMMER CAPACITOR (VARIABLE CAP.) or ADJUSTABLE/TUNABLE



This type of variable capac (a trimmer) is operated with small screwdriver or similar tool. It is designed to set when the circuit is made and then left without further adjustment.

DIODES;

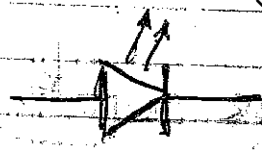
- DIODE

- It is a two terminal electrical device.
- They are made from a semiconductor, most often silicon but sometimes germanium.
- Types - Zener, rectifier, Transient voltage suppressor, thyristor, silicon controlled rectifier, TRIAC & Schottky, Laser, LED, Avalanche, PN-diode

This is a semi-conductor device which only allows current to flow in one direct.

- Step recovery diodes
- Constant current diode

- LED
LIGHT-EMITTING DIODE:

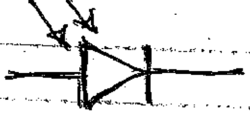


This is simply a transducer which converts electrical energy to light.

- ZENER DIODE;

- This is a special diode which is used to maintain a fixed voltage across its terminals.

- PHOTODIODE;



- A light sensitive diode.

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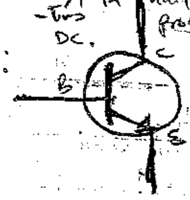
* Are diodes AC or DC ?
- Diodes conduct only when they are forward biased.
- Since, AC reverses its directn periodically, the diode conducts only in half cycles & insulates during the other cycles.
- This property of a diode is used in rectification of AC into DC.

TRANSISTOR:

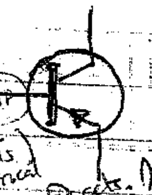
- TRANSISTOR (NPN):



This is simply a semi-conductor device i.e able to amplify or rectify an electric current.

It can be used with other components to make an amplifier or switching circuit.

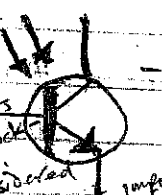
- TRANSISTOR (PNP)



Also a semi-conductor device that amplifies current or voltage in a cct. It can be used with other components to produce an amplifier.

* They are semi-conductor device used to amplify or switch electronic signals. Transistors are broadly divided into three (3) types: bipolar junction - BJTs (NPN/PNP)
- field effects (FETs)
- Insulated gate bipolar transistor (IGBTs)
- Device for amplifying, controlling & generating electrical signals.
- They are active components of integrated ccts. built with three (3) terminals for amplifying or switching cct.
- They are fundamentally an amplifier
- Acts as a switch & an amplifier, that is used to control or regulate the flow of electronic signals
- It's a miniature device that is used to
- Developed in 1947 by three American physicists John Bardeen, Walter Brattain & William Shockley. - It is considered as one of the most important inventions in the history of electronic or science.

- PHOTO TRANSISTOR



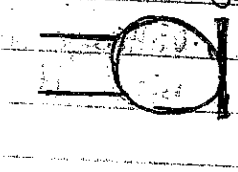
- This is a light-sensitive transistor.

It sense light levels & alter the current flowing b/w emitter & collector according to the level of light it receives.

- Base - is used to activate the transistor
- Collector - It is the positive lead of the transistor
- Emitter - is the negative lead.

AUDIO & RADIO DEVICES:

- MICROPHONE

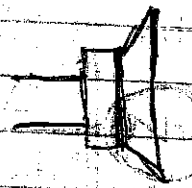


- A transducer which converts sound to electrical energy.

- EARPHONE

- A transducer which converts electrical energy to sound.

- LOUDSPEAKER

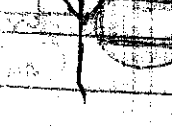


- A transducer which converts electrical energy to sound.

- PIEZO TRANSDUCER

- A transducer which converts electrical to sound energy.

- AERIAL (ANTENNA)

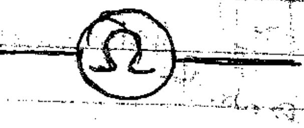


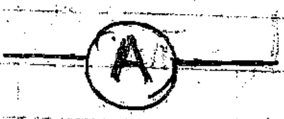
- A device which is designed to receive or transmit radio signals. It is also known as an Antenna.

COMPONENT	CCT SYMBOL	FUNCTIONS
- AMPLIFIER	I/P  O/P	An amplifier cct with one input. Really it is a block diagram symbol because it represents a cct rather than just one component.

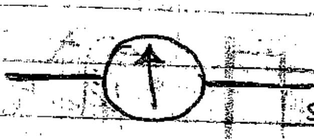
an Operational amp. is an Integrated cct that behaves like a high-gain difference amp. It amplifies the diff. b/w two input voltages. Op amp. is a triangle that has two inputs & a single output.
. An electronic device for increasing the amplitude of electrical signals, used chiefly in sound reproduction

METERS AND OSCILLOSCOPE:

- OHMMETER  An Ohmmeter is used to measure resistance. Most multi-meters have an ohmmeter setting

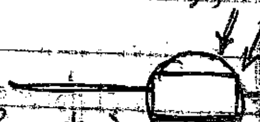
- AMMETER  ~~An~~ An Ammeter is used to measure current.

- VOLTMETER  A voltmeter is used to measure voltage. The proper name for voltage is 'Potential Difference' but most people prefer using voltage.

- GALVANOMETER  A Galvanometer is a very sensitive meter which is used to measure tiny currents, usually 1mA or less.

- OSCILLOSCOPE  An Oscilloscope is used to display the shape of electrical signals and it can be used to measure their voltage and time period.

SENSORS (INPUT DEVICES):

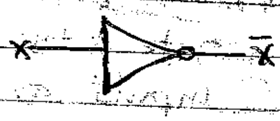
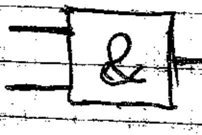
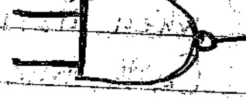
LDR =
(LIGHT DEPENDENT RESISTOR)  A transducer which converts brightness (light) to resistance (an electrical property).

THERMISTOR  A Transducer which Converts temperature (heat) to resistance (an electrical property)

LOGIC GATES

Basically Logic gates process signals which represents TRUE (1, high, +Vs, on) or FALSE (0, Low, 0V, off). Logic gate are physical devices That Can be used to Implement logic functions

There are two sets of symbols, i.e traditional and IEC (International Electrotechnical Commission).

GATE TYPE	TRADITIONAL SYMBOL	IEC SYMBOL	FUNCTION
NOT			A not gate is also Called an Inverter. It only have one input. The 'O' on The output means "not". The output of a NOT gate is the Inverse (opposite) of its input. So the output is True when the input is false.
- AND			These is a logic cct with more than one input terminals. No output signal is produced Unless a pulse is applied to all inputs Simultaneously. i.e If any of The Input are zero, the o/p will also be zero.
- NAND			This is a Combination of a NOT & AND function. It

TRUTH TABLE ⇒

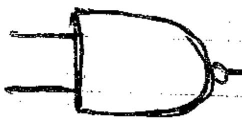
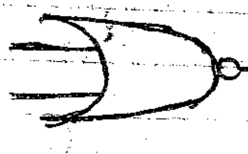
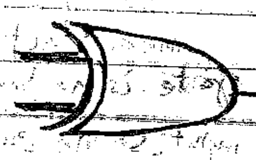
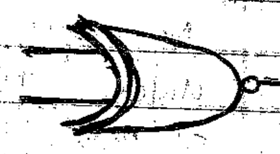
x	$\bar{x}$
1	0
0	1

Logical multiplication representation.

TRUTH TABLE:

x	y	z
0	0	0
0	1	0
1	0	0
1	1	1

Logical multiplication formation:

TYPE	TRAD. SYMBOL	IEC	FUNCTION
- NAND		&	It has two or more inputs and one output. The output is logic "0" on the output means 'not' showing that its output is true unless all its inputs are true.
- OR		≥1	An OR gate can have two or more inputs. The o/p of an OR gate is true when @ least one of its inputs is true.
- NOR		≥1	A NOR gate have two or more inputs. The 'o' on the o/p means 'not' meaning that it is a NOT OR gate. The o/p of a NOR gate is true when none of its inputs are true.
- Ex-OR		=1	An Ex-OR gate only have two inputs. The output of an Ex-OR gate is true when its inputs are different (one true, one false).
- Ex-NOR		=1	An Ex-NOR gate can only have two inputs. The 'o' on the output means "not" showing that it is a ~~NOT-Ex-OR~~ NOT EX-NOR GATE. The output of an EX-NOR gate is true when its inputs are the same (both true or both false).

ELECTRONIC COMPONENT IDENTIFICATION & INFORMATION

RESISTORS;

Standard resistors are "not" polarised and can be placed in the PCB any way around.

The most common resistor have four bands.

* Bands 1 and 2 - identify the first two digits, band 3 is the Multiplier and band 4 is the tolerance (normally Gold 5%)

The Multipliers are;

R = X0
K = X 1,000
M = X 1,000,000

The Kit parts list might contain for example, a 4K7 resistor. The K refers to the number multiplier (K = X 1,000). The position of the K between the 4 and 7 identifies the position of the decimal point so we have;

4·7 X 1,000 = 4,700 Ohms (with a colour code of Yellow, violet, Red)

Another example is say 56R. This gives us 56 X 0 = 56 Ohms resistor with colour code; Green, Blue, Black.

You will find, some five band resistors in our kits and the colour code is likely to be associated with the name precision 1% and 2% types. The work in a similar way to the four band but the first three bands are used to determine the numbers, band 4 is always the multiplier and band 5 is the tolerance.

See the table below for more details;

IDENTIFING COMPONENTS USING LETTERS

Components that are Commonly identified by letters are called Semi-conductor devices examples are diodes and transistors. It has not been extensively used for IC,S because of the difficulty in classifying the very great variety of I.C.s

The first letter indicates the type of Semi-conductor material used;

A – Germanium
B – Silicon
C – Gallium arsenide
D – Indium antimonide
R – photo Conductive Material. No Junction formed.

The Second letter indicates Construction or use of the devices;

A – Indicates that it is a detector diode, high speed diod Mixer diodes e.g BA154.
B – Variable Capacitance diode.
C – Transistor for Audio frequency application (not power output
D – Power transistor for audio frequency application
E – Means it is a tuned diode
F – Transistor for radio frequency
L – Power radio frequency

P- photo Sensitive devices or detector or radiated energy
Q- Radiation diode
R- Thyristor or Other Switching device (low power)
S- Switching transistor (Not power type)
T- Thyristor or Other Controllers (with High power)
U- Power transistor for Switching application.
X- Multiplier diode such as Varactor diode or varicapd.
Y- Rectifier diode.
Z- Voltage Reference or Regulator diode.

The Serial number which Consists of one letter U, X, Y, Z with two figures, simply means, It is a device for Professional used not available to amature, example, using BFY52,
BFY 57
ACY 17
BFX 84

The Serial number Consist of 3 figures intended primarily for used in Consumer electronic Such as radio, television, tape-recorder, audio-amplifier etc.

* THYRISTOR
Device that act as switches, rectifiers or voltage regulators

* In electronics, a varicap diode, varactor diode, variable capacitance diode, variable reactance diode or tuning diode is a type of diode designed to exploit the voltage-dependent capacitance of a reverse-biased p-n junction

* Def. The diode whose internal capacitance varies with the variation of the reverse voltage such type of diode is known as the varactor diode. It is used for storage of charge. It always works in reverse bias and it is a voltage dependent semiconductor device

* The varactor diode is used in a place where the variable capacitance is required, and that capacitance is controlled with the help

IDENTIFING COMPONENTS USING LETTERS

EUROPEAN TYPE;

The European type numbers gives us much more Information, and it Consist of 2 or 3 letters followed by a number Previously discussed. The first (1st) letter tell us the material used. for example A-for germanium or B- for silicon etc.
The Second (2nd) letter tell us what type of transistor we are working with or have, for instance, BC would indicate a silicon general purpose transistor etc.
The Third (3rd) letter is often Used when the transistor has been developed for Industrial use, as distinct from entertainment. The number next which follow indicates the stage of development A high number means a recent type device.

AMERICAN TYPE;

American type Consist of a prefix 2N, 1N, followed by a Serial number for the given time. 2N Signifies that the device is a transistor. 1N means it's a diode, the number been the registration number means nothing e.g 1N4148, diode, 2N 3055 - power transistor, and 2N4044 - thyristor.

JAPANESE TYPE;

Japanese numbering always indicated with 2S followed by a letter A, B, I, E, or D and a Serial number of 3 or 4 figure e.g
1.) All 2S, AS and 2SBS Series are PNP transistors

2) All 2SE, 2SD series are NPN transistor. Note that all actual type number printed on the transistor Omitted the 2S. Also note that, a lot of transistors are now available in the Market with the type number already described example; OK870, GET 12, NKT 713, MJE 205, OC 71, TIP 30, 31 etc

Note; 2N169 is MOS FET Transistor.

Tip 29 is a silicon power transistor PNP

Tip 30 is a Silicon Power transistor NPN.

COMPONENTS COMMONLY USED IN ELECTRONIC CIRCUIT

RESISTORS: Resistors in an electronic Circuit are Used for Many purposes but in all Cases their property is to restrict the flow of Current in the Circuit. Every electronics Project Used a number of resistor and they are probably used in larger quantities than any other type of Component.

The value of the device is usually shown by a system of color coding, with resistors of this nature having four color bands around its bodies. The first two bands indicate the first two digits of the value and the third band is the multiplier. The fourth band indicates the tolerance of the Component. If occasionally you Come across a fifth color band. The first band will be Salmon Pink in color, this indicate that the color. is highly stability type.

Another form of resistor coding is sometimed used, this consist of combination of letters and numbers which are marked on the body of the component. Examples are;

1R0J = 10 Ohms 5%	5R6K = 5.6 Ohms 10%
10RF = 10 Ohms	56RJ = 56 Ohms 50%
K10K = 100 Ohms 10%	K56J = 560 Ohms 5%
1K0J = 1K Ohms 5%	5K6G = 5.6 K Ohms 2%
10KG = 10K ohms 2%	56KM = 56K Ohms 20%
IM0F = 1M	M56K = 560K Ohms 10%
10MK = 10 Ohms 10%	5M6M = 5.6M Ohms 20%

The last letter is always the tolerance as followed below;

F = ±1%,

G = ±2%

J = ±5%

K = ±10%.

M = ±20%.

CAPACITORS;

This are usually described by their dielectric and may vary in Capacity from a few pico-farads to many thousands of Micro-farads, and have voltage ratings from a few volts to many Kilovolts. Despite all these variations, the device, Capacitor in cct diagram are designated by one symbol, unless they are polarised e.g (electrolytic).

The value of Capacitors are express in farads, and its unit is named after Faraday. Capacitor values are normally expressed in terms of Micro-farads, Nano-farads or Pico-farads. A thousand of Micro-farad, and a Pico-farad is equal to one thousand of a nano-farad. Micro-farad is abbreviation as M while Micro-farad Mfd or Nano-N. Pico is P. Also Capacitor value are often expressed in terms of NF or Pf.

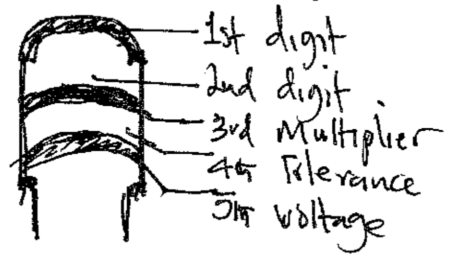


fig a.) C280 TYPE CAPACITOR COLOUR CODING

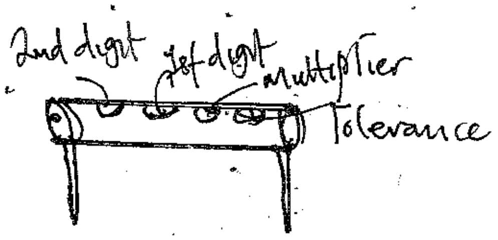


fig c. TUBULAR CERAMIC COLOUR CODING

fig b.) DISC CERAMIC COLOUR CODING

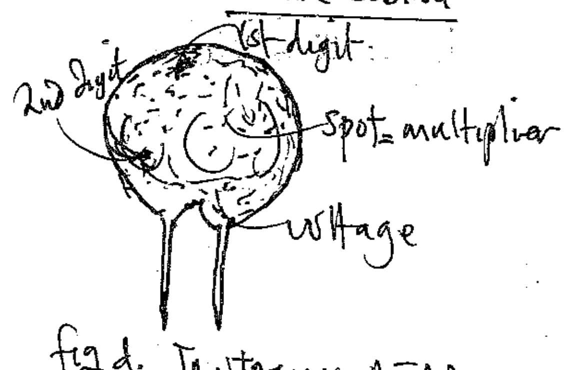
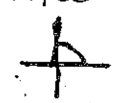


fig d. TANTALUM BEAD COLOUR CODING

The element shown in a Schematic are refer to by letters and for numbers. The letters are usually abbreviated from a Complete name e.g resistor are marked by letter R, and Capacitors are C etc.

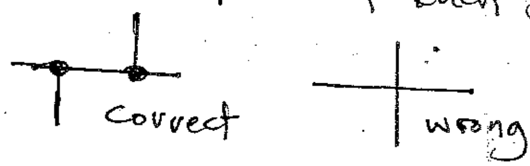
Ordinarily paths for signal or energy are shown from start to end. If however lines may be broken off (as it is usually done for filament Circuits) At the break, the line is terminated in an arrow head and bears an inscription starting its destination. So that a Schematic will not look too Crowded. Groups of lines may be shown by a single line If an electronic System has several sub-system or devices, it is usually easy to draw-up a seperate schematic for each of them.

GUIDE LINE FOR DRAWING A WELL ARRANGED SCHEMATIC DIAGRAM;

1.) When Conductors ~~must~~ are Crosses in a schematic diagram, they appropriately do so at the right angle 

2.) When one Conductor Joins another, the joint must be indicated by a dot drawn with a diameter of about 1mm. If The Joint is a T-Joint, at one point only. Meaning that only one Dot must ~~be~~ used. 

3.) If the Joint is double-T (or Cross Joint), the end of the lines (representing the Joining Conductors) are displaced side ways at one angle 45° at a distance of about 3mm and a dot is placed at point of each Joint.



4.) Connection to Circuit Components are always made a small distance from the components symbols and not right at the Component Symbol.

5.) The Main rule in the layout of a Schematic diagram is that signal flow must be from left to right or from top to bottom (or a Combination of both if possible).

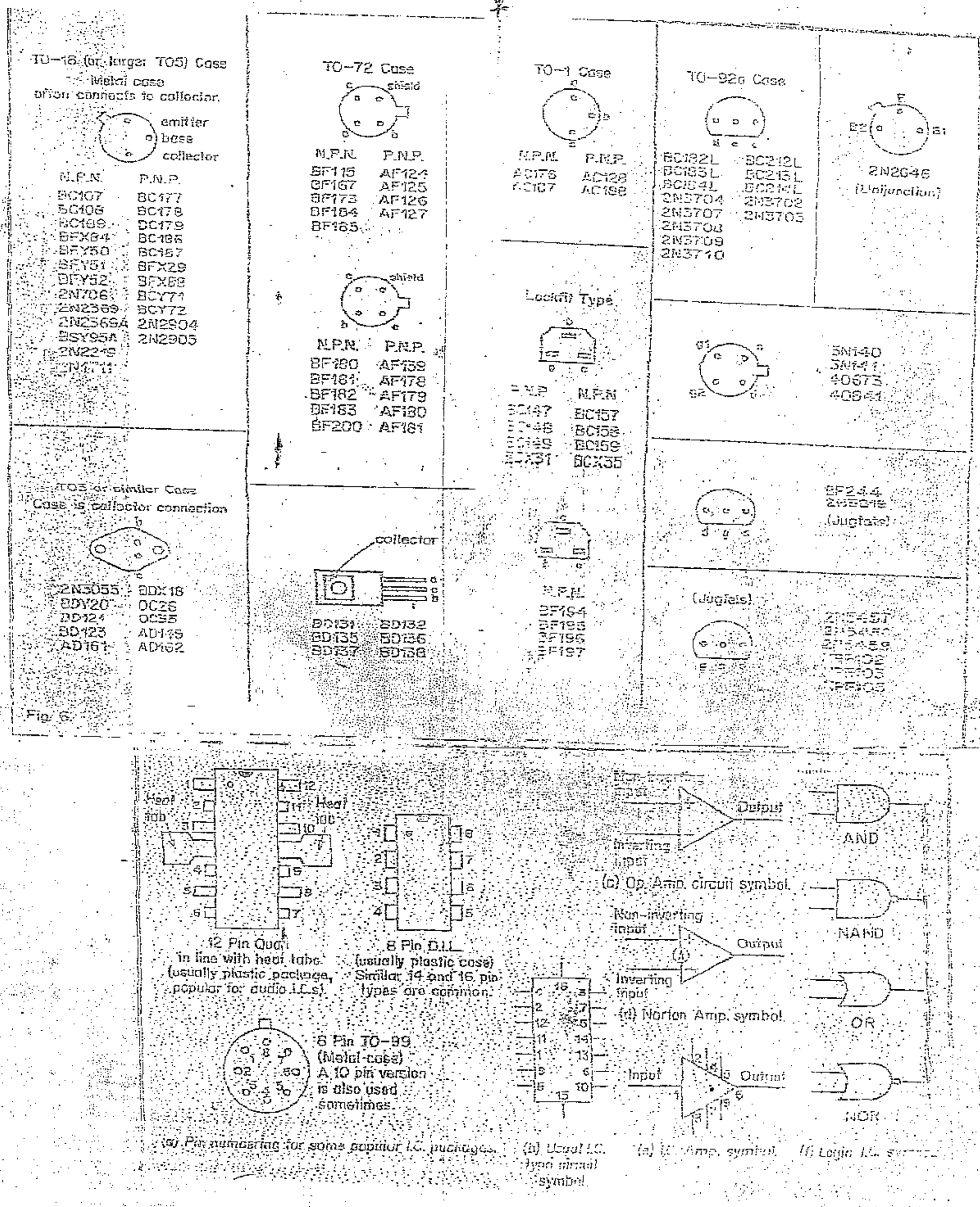
6.) Understand the functional Assembly and identify each Component that makes up the Circuitry.

7.) Ensure that all Components on the Circuit are included in the Schematic diagram and well fixed in their proper location, and they are all lettered & numbered.

8.) Avoid usage of different symbols for the same device.

9.) Transistor and tube symbols should be aligned horizontally.

EE437 (ELECTRONIC DEVICES)



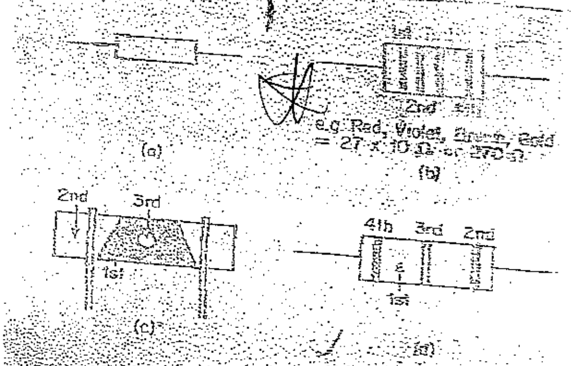
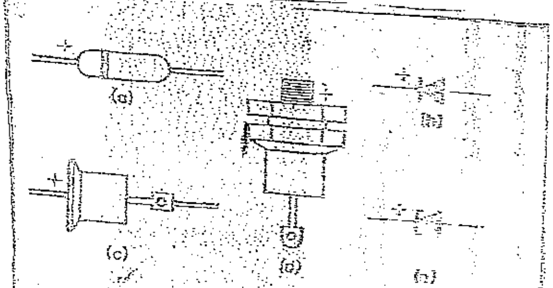


Fig. 1. (a) Resistor circuit symbol, (b) ordinary resistor colour coding, (c) and (d) coding sometimes used for resistors having high power ratings.



4. (a) A small diode, (b) diode symbol, (c) Top Hat type rectifier, (d) Stud type rectifier, (e) Zener circuit symbol.

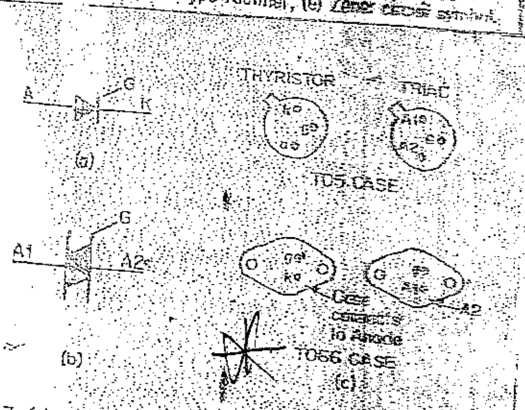


Fig. 7. (a) Thyristor circuit symbol, (b) Triac circuit symbol, (c) connection data.

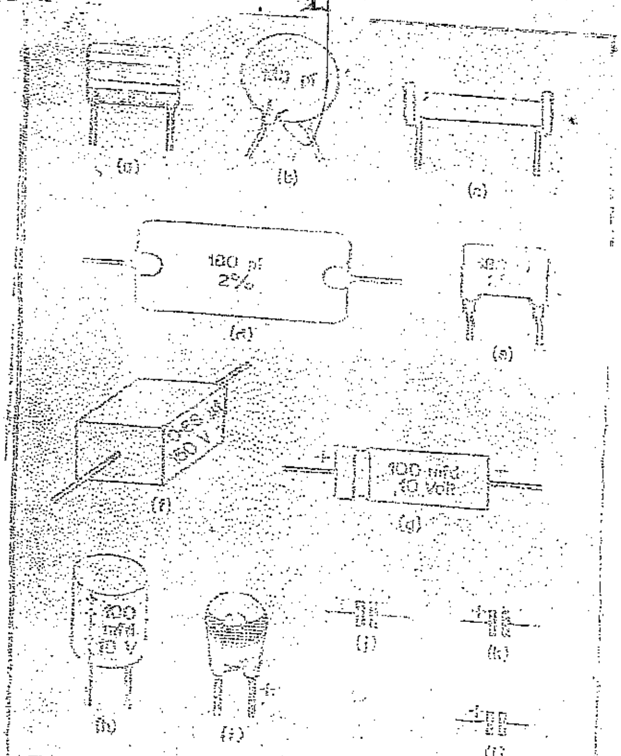


Fig. 2. (a) Axial lead capacitor, (b) a disc ceramic, (c) a tubular ceramic, (d) a silvered mica type, (e) a moulded silvered mica type, (f) some high value non-electrolytic types take this form, (g) an electrolytic, (h) vertical (p.c.b.) type electrolytic, (i) a tantalum bead, (j) capacitor (non-electrolytic) circuit symbol, (k) electrolytic circuit symbol, (l) alternative for (k).

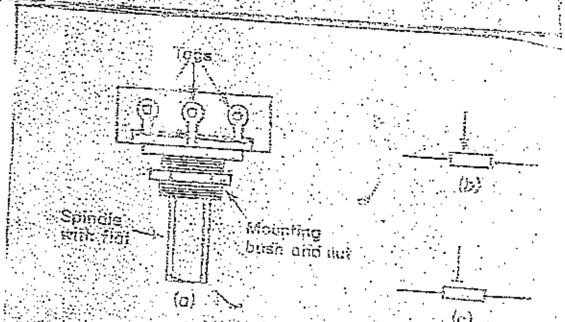


Fig. 6. (a) A potentiometer, (b) its circuit symbol, (c) preset circuit symbol.

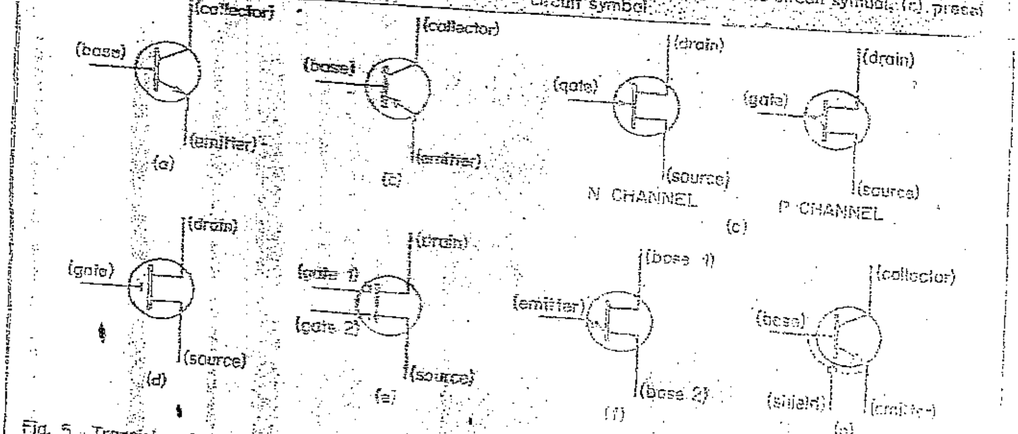


Fig. 5. Transistor circuit symbols: (a) n.p.n. bipolar, (b) p.n.p. bipolar, (c) N and P channel junction ..., (d) MOSFET, (e) dual gate MOSFET, (f) unijunction, (g) bipolar (n.p.n.) with shield.

FORMS DEFINITION & APPLICATION OF DIAGRAMS

BLOCK DIAGRAM;

DEFINITION: This is simply defined as a diagram of a System, instrument which selected portion are represented by a given annotated boxes and interconnected ~~wiring~~ lines or wires.

A Block diagram is a graphical representation of a System, which provides a functional view of the system, ~~and also~~ gives a better understanding of a system's functions and also help Create interconnections within it. Block diagrams derive their name from the rectangular elements found in this type of diagram. They are used to describe hardware and software Systems as well as to represent Processes.

Block diagram are described and defined according to their function and structure as well as their relationship with other blocks. The function performed by each unit is stated in verbal form inside the block. The relationship to one another is indicated by appropriately Connected lines and arrows, the arrow-heads point from the Controlling I/P signal towards the Controlled Unit.

WHY ARE BLOCK DIAGRAMS IMPORTANT?

A block diagram is an essential method used to developed and describe hardware or software Systems as well as represent their workflows and processes.

- They are used in electronics to represent systems in the trucking industry.
- Block diagrams are generally used when the visualization of Information or Control flows is important, or when processes are involved. In this way, we Can represent Complex algorithms or flows ~~Components within a large System, as with~~ of information or Communication among Individuals Components within a large system as with it. For example In a facility designed for Mass production.

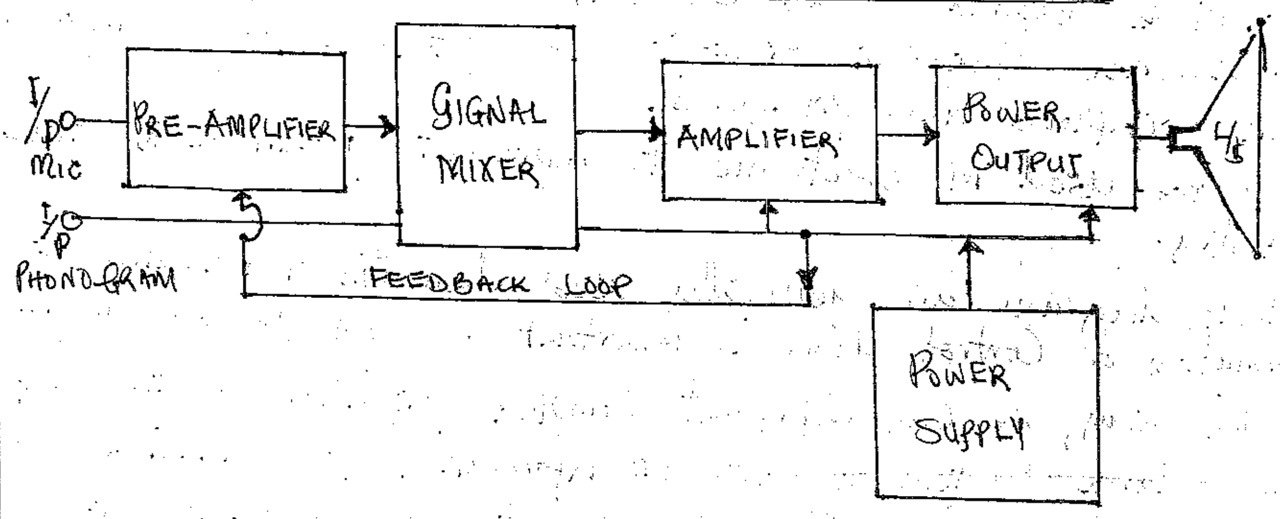
STEPS TO TAKE IN DRAFTING BLOCK DIAGRAM:

1- Understand the functional Assembly of the Circuitory.
2- Identify the various stages involved.
3- Study the signal flow "left to right" or from "top to bottom"
4- The input and output must be shown or must be determined.
5- All the sub-connections from feedback loops must be clearly identified and indicated.
6- The engineer designer usually begins the design of an electronic devices by laying out a block-diagram. This block diagram is then used as a guide while the designs Circuit represented by varies blocks in the system.

APPLICATION: In design work, block diagrams are drawn up prior to all other forms. In service, they give an over all idea about the operation of the equipment.

To illustrate this, Consider the block diagram of an audio amplifier, assume the input signal will be from Microphone or phonogram, pick-up, the o/p signal must be large enough to activated a loudspeaker. A typical diagram of this device can be sketched as shown below;

BLOCK DIAGRAM OF AN AUDIO AMPLIFIER



SCHEMATIC DIAGRAM (ELEMENTARY); 34

A schematic diagram is a visual representation of the elements of a system (ccts) using abstract, graphic symbols rather than realistic pictures. It is defined as a picture that represents the components of a process, device or other object using abstract, often standardized symbols and lines.

A schematic diagram is basically a fundamental two dimensional circuit representation showing the functionality and connectivity b/w different electrical components. It is vital for a PCB designer to get familiarized with the schematic symbols that represent the components on a schematic diagram. Although schemetic diagrams are commonly associated with electrical ccts, many examples can be found in other industries. Schematic diagram only "depict" the significant components of a systems, though some details in the diagrams may also be exaggerated or introduced to facilitate the understanding of the system.

- They do not include details that are not necessary for comprehending the information that the diagram was intended to convey. For example, in a schematic diagram depicting an electrical cct, you can see how the wires and components are connected together, but not photographs of the circuit itself.

- More simply a schematic diagram is a simplified drawing that uses symbols and lines to convey important information and they are most commonly associated with electronic components in electrical circuit and they are also called wiring diagram or cct diagrams.

STANDARDS FOR SCHEMATIC SYMBOLS

IEC 60617: International Electrotechnical Commission, has issued this standard. It is based on the older standard. British standard BS 3939. This database includes over 1750 schematic symbols.

ANSI STANDARD Y32: American National standard institude (ANSI) This provide a variety of specialized symbols, originally used for aircraft applications. A series of minor changes performes on this standard has made the existing document aligned with IEC.

APPLICATION:

A schematic (elementary) diagram serves as a basic for other diagrams and drawings. It gives an insight in operation of individual circuit and components. Useful in testing, repair and alignment and services.

* SIGNAL FLOW (FUNCTIONAL DIAGRAM)

In functional diagram, a diagram tracing the passage of valid signals in an equipment diagram or part there-of.

APPLICATION!

A signal flow diagram gives an insight to operation principle of a system. It is also useful in the testing, alignment and repair of the equipment or instrument.

* WIRING (INTERNAL CONNECTION DIAGRAM)

This is simply a diagram that shows the internal connection of a system and / with details and accessory items such as cable or wire, type terminals blocks, connection leads etc.

APPLICATION! This form of diagram is used when putting a system or equipment into services.

* SYSTEM DIAGRAM:

This is basically a diagram that relates the sub-system of a system according to functions they perform in services.

APPLICATION: A system diagram is used to get a general idea about the sub-systems and a system as a whole during installation, inspection or operation.

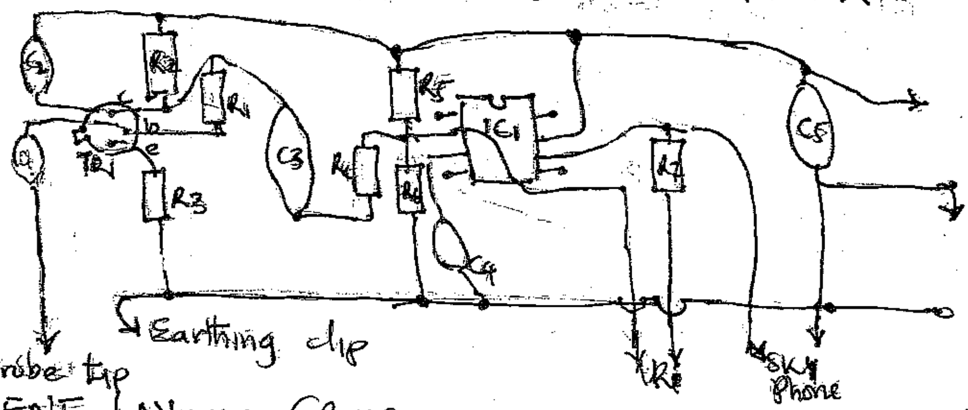
* LAYOUT DIAGRAM:

This is a diagram that shows all the components of an equipment in their physical relation to one another.

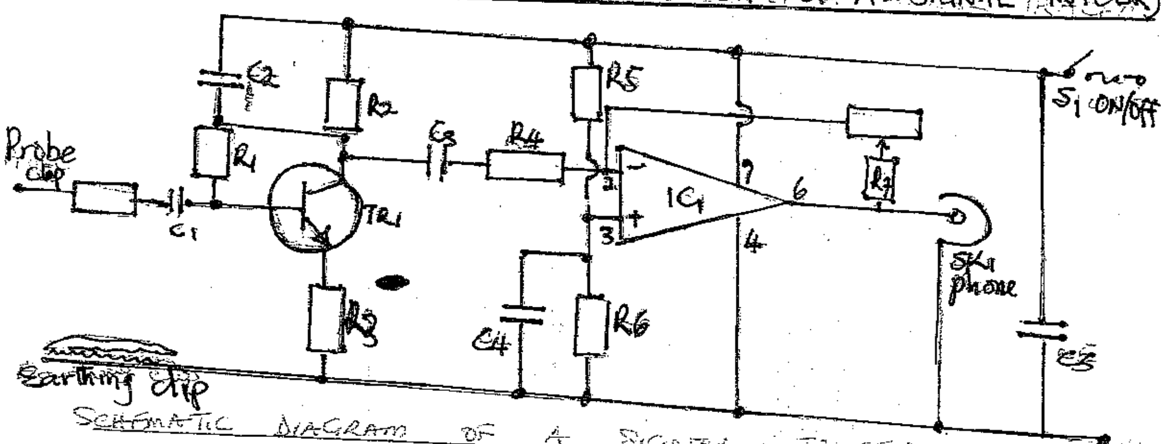
APPLICATION!

A Layout diagram is used in working out engineering drawings and also when installing or maintaining an equipment.

SCHEMATIC DIAGRAM OF A SIGNAL TRACER



COMPONENTS LAYOUT (ROUGH SKETCH OF A SIGNAL TRACER)



SCHEMATIC DIAGRAM OF A SIGNAL TRACER

Components List of the above diagram.

R1	3m3
R2	10K
R3	1K
R4	22K
R5	39K
R6	39K
R7	22K
C1	.22nF
C2	10nf
C3	20.0 nf
C4	100 nf
C5	100 nf.
VR1	2m2 lin
IC1	LF 351.
TR1	BC 109C
S1	Single Pole (SPST).

* External Connection Diagram

This diagram simply shows the external connections of a system or equipment and contain details and accessories. Similar to these in an internal connecting diagram.

APPLICATION: This form of diagram is used when putting a system or equipment into service.

* Logic Diagram;

Logic diagram is a diagram representing all the logical elements and their interconnection without necessarily expressing constructing or engineering details.

In Logic diagrams, every symbols represent a unique logical device, It gives more idea about the circuit more than a block diagram. Each straight line in a logic represent a conductor.

In design work, block diagram is drawn prior to all other forms in service, they basically give an overall idea about the operation of the equipment.

Procedure/Steps in Drafting Logic Diagram

- Project or have a sketch of the schematic with all the symbols
- You should be able to recall what each symbols means
- You should be able to know dimension for each symbols
- Note that signal flow from left to right.
- Start by drawing input ward matrixes.
- Should tie the devices to the input terminals.
- Identify the number of input and output required.

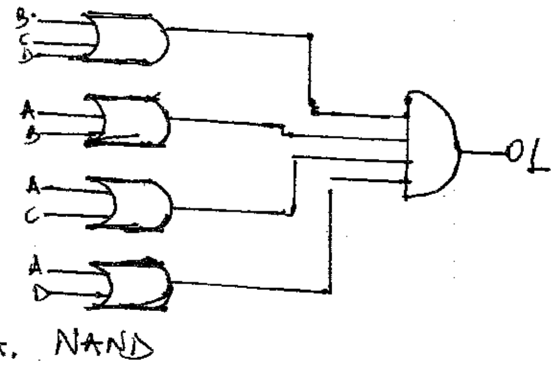


fig A. NAND

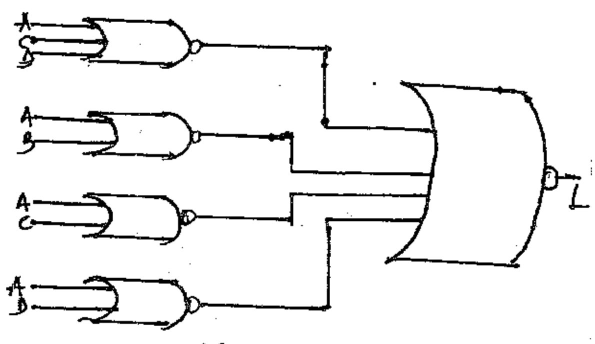


fig B. NOR

LOGIC DIAGRAMS;

WHAT'S ELECTRICAL CIRCUIT-BOARD?

A cct board is a physical piece of technology that allows for the assembly of electrical or data ccts on a horizontal layer of Material. It is the heart of the electronic Circuitory that controls the Circuitory of the device and provides Power to the device.

Circuit Construction

Types of Circuit Board

* **Solderless Breadboard:** This type of circuit board is used temporary, which therefore required "no soldering" in its process or in used.

This is a way of making a temporary circuit, for testing purposes or to try out an idea. No soldering is required and all the components can be re-used afterwards. It is easy to change connections and replace components. Almost all the Electronics club projects started life on a breadboard to check that the circuit worked as intended.

Uses of Breadboard:

Solderless breadboard is basically used in the laboratories for low powered project as preparation board so as to confirm that the project is effectively working before transfering it or using the permanent board.

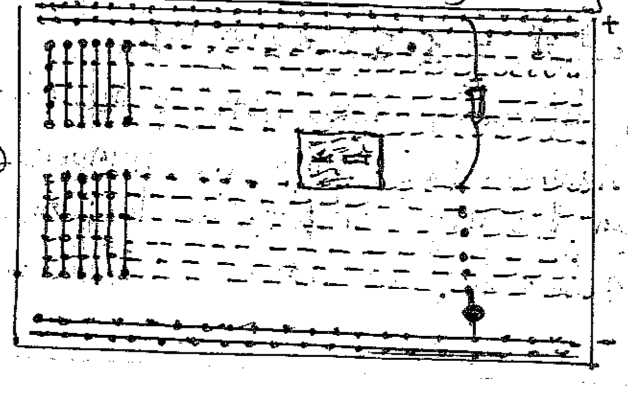
As it is called, this board needs not to be soldered, because clips are attached below each hole as directed by the manufactures, base on the fact that is used to make-up temporary circuits for testing or to try out an idea. Parts will not be damaged so they will be available to re-used afterwards.

Connections on Breadboard

Breadboards have many tiny sockets (called "holes") arranged on 0.1" grid. The leads of most components can be pushed straight into the holes. ICs are inserted across the central gap with their notch or dot to the left.

-Wire links: Can be made single-core plastic-coated wire of 0.6mm diameter (the standard size). Stranded wire is not suitable because it will crumple when pushed into a hole and it may damage the board if strands break off.

The diagram shows how the breadboard holes are connected



The top and bottom rows are linked, horizontally all the way across as shown by the red and black lines on the diagram. The power supply is connected to these rows, + at the top and 0v (Zero volts) at the bottom.

I suggest, using the upper row of the bottom pair for 0v, then you can used the lower row for the negative supply with circuits requireng a dual supply (example +9v, 0v, −9v). The other holes are linked "Vertically" in blocks of 5 with no link across the centre as shown above, by the blue lines on the diagram. Notice how there are seperate blocks of connections to each pin of ICs.

LARGE BREADBOARDS

On Larger breadboard, there may be a break halfway along the top and bottom power supply rows. It is a good idea to link to link across the gap before you start building the circuit, otherwise, you may forget and part of your cct will have no power.

BUILDING A CIRCUIT ON BREADBOARD

Converting a circuit diagram to a breadboard layout is not straightforward because the arrangement of components on breadboard will look quite different from the circuit diagram.

When putting parts on breadboard you must concentrate on their connections, not their positions on the circuit diagram. The IC (chip) is a good starting point, so placed it in the centre of the breadboard and work round it pin by pin, putting in all the connections and components for each pin in turn.

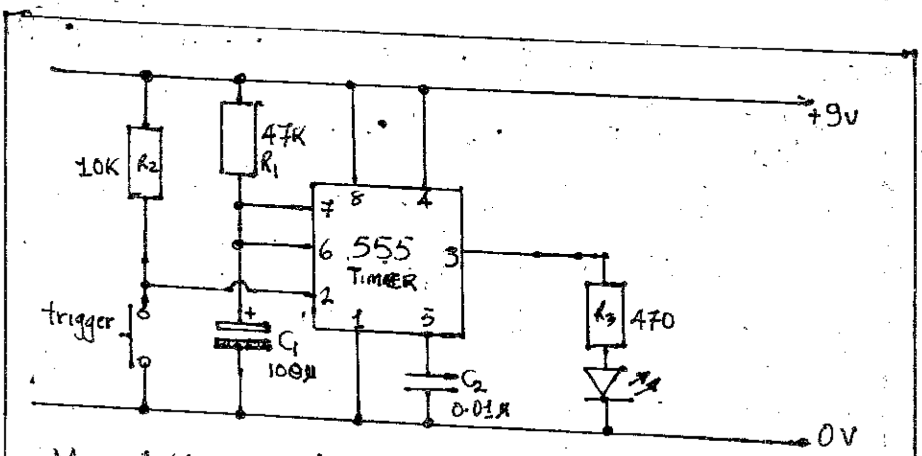
The best way to explain this is by these example, so the process of building this 555 timer circuit on breadboard is listed step by step below.

The circuit is a monostable which means, it will turn on the LED for about 5 seconds when the "trigger" button is pressed. The time period is determined by R_1 and C_1 and you may wish to try changing their values. R_1 should be in the range 1KΩ to 1MΩ.

Time Period (T) = $1.1 \times R_1 \times C_1$

For further information, take note of the 555 Monostable.

IC Pin Numbers: IC pins are numbered anti-clockwise around the IC starting near the notch or dot. The diagrams shows the numbering for 8pin and 14pin ICs, but the principle is the same for all sizes.



Monostable Circuit Diagram

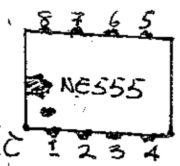


fig. 8pin IC

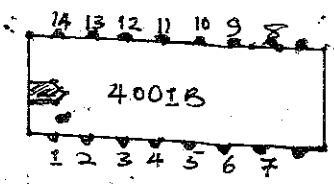
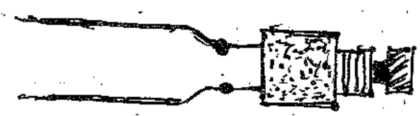


fig kiis 14 pin IC

COMPONENTS WITHOUT SUITABLE LEADS

Some Components such as switches and variable resistors do not have suitable leads of Their own, so you must solder some on yourself.



Use single-core plastic coated wire of 0.6mm diameter (the standard size). Stranded wire is not suitable because it will crumple when pushed into a hole and it may damage the board if strands break off.

IMPORTANT KEYNOTE;

- Check all the Connections Carefully
- Check that parts are connected correctly way round (LED and 100µF capacitor)
- Check that no leads are touching (unless they connect to the same block)
- Connect the breadboard to a 9v supply and press the push switch to test the circuit.
- If your Circuit does not work as required disconnect (or switch off) the power supply and very Carefully re-check every Connection against the Circuit diagram.

Solderless breadboard usually cannot accommode "SURFACE-MOUNT TECHNOLOGY" device (SMD) or non 0.1" (2.54mm) grid spaced components, like for example those with 2mm spacing. Further, they can not accommodate components with multiple rows of connectors, if these connectors don't match the DIL Layout (its impossible to provide correct electrical connectivity). Sometimes small PCB adapters (breakout adapters) can be used to fit the component on such adapters. Carry one or more of the non-fitting components and 0.1" (2.54mm) connectors in DIL Layout. ~~A~~ Larger ~~of the~~ components are usually plugged into a socket, where the socket was soldered onto such an adapter. The smaller components (e.g SMD resistors) are usually directly soldered onto such as adapter. The adapter is then plugged into the breadboard via the 0.1" connectors. However, the need to soldered the component or socket onto the adapter contradicts the idea of using a solderless breadboard for prototyping in the first place.
Therefore complex circuits can become unmanageable on a breadboard due to the large amount of wiring necessarity.

PLAIN MATRIX BOARD OR PLAIN VERO BOARD

Plain matrix board is often referred to as plain vero board which is the trade name of vero-electronics LTD.
However, it is produced by other manufacturers; matrix board which consist of a sheet of insulated material (resin boarded paper) which has been divided with a matrix of small holes. These boards are available in various sizes and in two pitches i.e (2.54m = 1mm and 3.8mm (0.15m) the 3.8mm (0.15m) matrix type is for general purpose or used. The reason for this is that the most popular ICs have pins that are spaced on a 2.5mm.
To suit a particular circuit and component layout, so that, they do not have a matrix of holes, these are only drilled where components are actually going to be mounted.

PLAIN MATRIX BOARD LAYOUT

* Almost invariable the Components are mounted on the matrix panel using the same layout as they used in the circuit diagram.
* You can therefore keep the practical layout as much like the cct layout as possible, or can make a few simple modifications which will provide a more compact layout without any crawping of components but avoid the necessity to insulate any of them with PVC sleeving. This is usually the case of the physical layout been kept as much the same as the theoretical one, but not likely to be the case of the physical layout, been nothing compare like the circuit
* The normal way of mounting a circuit board is to bolt it to the case of the equipment and for small panel, it is usually suff icient to use two mounting points.
* Note that, three rows of used holes should be left out to provide space for these holes.

PLAIN MATRIX BOARD CONSTRUCTION

By construction, firstly cut out a panel of the correct size. It is easier to cut along a row of holes rather than between rows.

- A Junior hack-saw or a full size type can be used for that process.
- Any mounting holes can be drilled and in-case they can be made using a 3.2mm drill-bit, so that they will accept m3 mounting bolts.
- Next, the components that are mounted on the panel, their lead out are bent over at right angle on the underside of the board, they ~~are~~ should be wired together in their correct method.
- Connect first the component to the negative supply rail (or positive supply rail in case of positive earth equipment) before wiring up the rail, one by one. The other components are mounted and soldered into circuit.
- Another method is to start off mounting all the components and then cut their lead out, wires to length and solder them together as necessary.

POINT-TO-POINT SOLDERING

The process of soldering simply the end of lead-out-wires (or tags) etc, with solder which also simply known as TINNING. When point-to-point wiring is used (Lead and tags are soldered directly to one another. It is advisable to tin all tags and leads before attempting to complete any Joints. Though this tinning is less important on printed circuit board unless a lead or tag is badly Contaminated whatever method is used; always make sure that plenty of solder is used on each Joint when carrying out point-to-point wiring.

PRINTED CIRCUIT BOARD

Printed circuit boards basically Consist of thin sheet of an insulation material such as S.R.B.P or Fiber-glass, and on one side of the board, the present of copper exist. PCB have copper tracks connecting the holes where the components are placed. They are designed specially for each circuit and make Construction very easy. However, producing the P.CB requires special equipment so this method is not recommended if you are a beginner unless the P.CB is provided for you. The component's are mounted on the non-coppered side of the board with their load passing through the holes. The leads are cut to length on the side of the board, so that 1mm to 2mm of lead-out is left, protruding through the panel, the Leads are then soldered to the copper backing.

The purpose of the copper backing is of Course to Connect components together in their appropriate or proper-way.

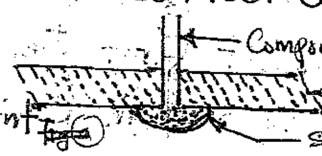


Fig PCB1 - A good PCB soldering Joint

The above diagram shows a side view of a good PCB soldered Joint. Here the soldered flowed tightly around the lead and over a fairly large area of copper backing. A part from providing a good electronic Connection, it also provide a firm physical mounting for the components which is very important.